

# Physics 1 Test

Chapters 3 and 4: Forces and motion

|             |                                                 |
|-------------|-------------------------------------------------|
| <b>Name</b> | _____                                           |
| <b>Date</b> | Thursday 15 October 2026                        |
| <b>Time</b> | 90 minutes                                      |
| <b>Aids</b> | Part 1: none. Part 2: all, except communication |

---

Answer all the exercises. Show your working and justify your answers; a correct answer without justification does not earn full credit. Use figures where helpful.

Part 1 is handed in before you begin Part 2. The test has 5 exercises and is worth 20 points in total.

## Part 1

Without aids

### Exercise 1 – Forces on an incline (4 points)

A block of mass  $m$  rests on an incline making an angle  $\theta$  with the horizontal.

- Draw a figure and mark all the forces acting on the block.
- Show that the normal force is  $N = mg \cos \theta$ .
- How large must the friction force be, at minimum, for the block to stay at rest?

### Exercise 2 – Equations of motion (4 points)

A ball is thrown straight up with speed  $v_0 = 12 \text{ m/s}$ . Neglect air resistance, and use  $g = 9.8 \text{ m/s}^2$ .

- How high does the ball go?
- How long does it take before the ball is back at its starting point?

### Exercise 3 – Vectors (2 points)

Two forces  $\vec{F}_1$  and  $\vec{F}_2$  act at the same point.  $\vec{F}_1$  has magnitude 3.0 N and points east;  $\vec{F}_2$  has magnitude 4.0 N and points north. Find the magnitude and direction of the resultant force.

## Part 2

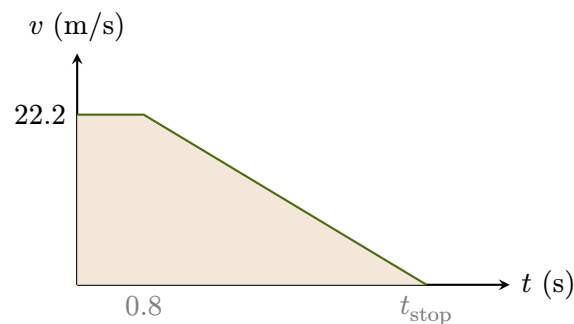
With aids

### Exercise 4 – The braking car (6 points)

A car is travelling at 80 km/h when the driver notices an obstacle 50 m ahead. The reaction time is 0.8 s, and the brakes give a constant deceleration of  $6.5 \text{ m/s}^2$ .

- How far does the car travel during the reaction time?
- Does the car manage to stop before the obstacle? Show your working.

The figure below shows a velocity–time graph of the motion.



**Figure 1.** Velocity as a function of time. The area under the graph is the distance.

- Use figure 1 to explain how the stopping distance can be read off the graph.

### Exercise 5 – Numerical solution (4 points)

The program below simulates the fall of a ball with air resistance using Euler's method.

```
1 import numpy as np
2
3 m, g, k = 0.10, 9.81, 0.02      # mass (kg), g, air resistance
4 v, t, dt = 0.0, 0.0, 0.01
5
6 while t < 5.0:
7     a = g - (k / m) * v**2      # acceleration (m/s^2)
8     v = v + a * dt
9     t = t + dt
10
11 print(f"Speed after {t:.1f} s: {v:.2f} m/s")
```

- a) Explain what line 7 does, and which physical law it is based on.
- b) What will the program print if  $k$  is set to zero? Justify your answer without running the program.
- c) Change the program so that it also computes how far the ball has fallen.

---

Good luck!